

R4 Newton's third law

Name: _____

Class: _____

Date: _____

Time: **408 minutes**

Marks: **339 marks**

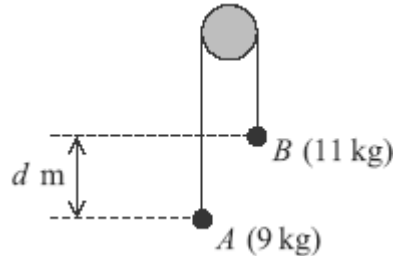
Comments:

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q1.

Two particles, A and B , are connected by a string that passes over a fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A .

The motion of the particles is to be modelled using simple assumptions.

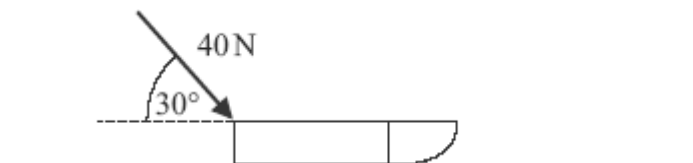
- (a) State one assumption that should be made about the peg. (1)
- (b) State two assumptions that should be made about the string. (2)
- (c) By forming an equation of motion for each of the particles A and B , show that the acceleration of each particle has magnitude 0.98 m s^{-2} . (5)
- (d) When the particles have been moving for 0.5 seconds, they are at the same level.
 - (i) Find the speed of the particles at this time. (2)
 - (ii) Find d . (4)

©2025 Exam Papers Practice. All rights reserved.

(Total 14 marks)

Q2.

A sledge of mass 8 kg is at rest on a rough horizontal surface. A child tries to move the sledge by pushing it with a pole, as shown in the diagram, but the sledge **does not move**. The pole is at an angle of 30° to the horizontal and exerts a force of 40 newtons on the sledge.

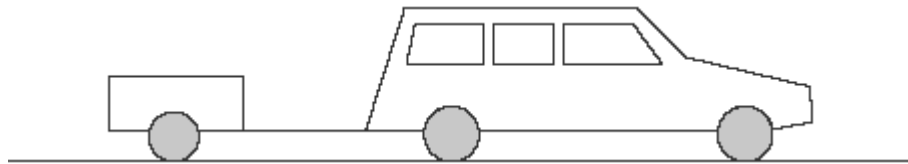


Model the sledge as a particle.

- (a) Draw a diagram to show the four forces acting on the sledge. (1)
- (b) Show that the normal reaction force between the sledge and the surface has magnitude 98.4 N. (3)
- (c) Find the magnitude of the friction force that acts on the sledge. (2)
- (d) Find the least possible value of the coefficient of friction between the sledge and the surface. (3)
- (Total 9 marks)**

Q3.

A car, of mass 1400 kg, is towing a trailer, of mass 600 kg. The two vehicles accelerate together at 1.3 m s^{-2} along a straight horizontal road.



- (a) Find the distance that the car and trailer would travel while accelerating from rest to 13 m s^{-1} . (3)
- (b) A forward driving force, of magnitude 3900 N, acts on the car. A resistance force, of magnitude 800 N, also acts on the car. (3)
- (i) A resistance force, of magnitude P newtons, acts on the trailer. Find P . (3)
- (ii) Find the magnitude of the force that the car exerts on the trailer. (3)
- (Total 9 marks)**

Q4.

A block, of mass 14 kg, is held at rest on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.25. A light inextensible string, which passes over a fixed smooth peg, is attached to the block. The other end of the string is attached to a particle, of mass 6 kg, which is hanging at rest.



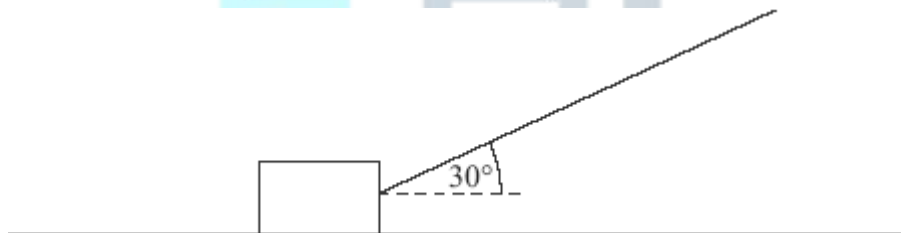
The block is released and begins to accelerate.

- (a) Find the magnitude of the friction force acting on the block. (3)
- (b) By forming two equations of motion, one for the block and one for the particle, show that the magnitude of the acceleration of the block and the particle is 1.225 m s^{-2} . (5)
- (c) Find the tension in the string. (2)
- (d) When the block is released, it is 0.8 metres from the peg. Find the speed of the block when it hits the peg. (3)
- (e) When the block reaches the peg, the string breaks and the particle falls a further 0.5 metres to the ground. Find the speed of the particle when it hits the ground. (3)

(Total 16 marks)

Q5.

The diagram shows a block, of mass 20 kg, being pulled along a rough horizontal surface by a rope inclined at an angle of 30° to the horizontal.



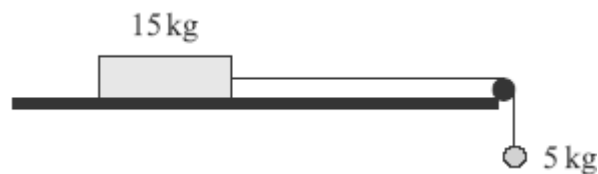
The coefficient of friction between the block and the surface is μ . Model the block as a particle which slides on the surface.

- (a) If the tension in the rope is 60 newtons, the block moves at a constant speed.
 - (i) Show that the magnitude of the normal reaction force acting on the block is 166 N. (3)
 - (ii) Find μ . (4)
- (b) If the rope remains at the same angle and the block accelerates at 0.8 m s^{-2} , find the tension in the rope. (5)

(Total 12 marks)

Q6.

A block, of mass 15 kg, is placed on a rough horizontal surface. It is attached, by a light inextensible string that passes over a smooth peg, to a particle of mass 5 kg, which hangs freely, as shown in the diagram.



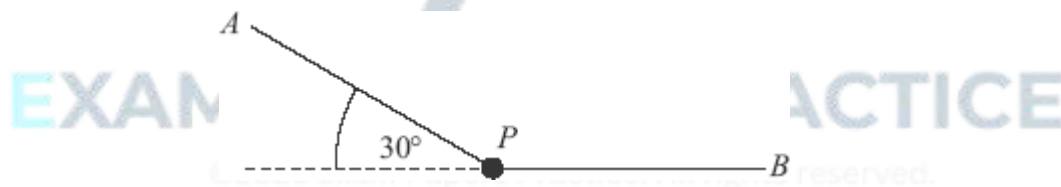
The coefficient of friction between the block and the surface is 0.2. The block and particle are released from rest and move with the string taut.

- (a) Find the magnitude of the friction force acting on the block. (3)
- (b) Show that the acceleration of the system is 0.98 m s^{-2} . (5)
- (c) The block is travelling at 1.4 m s^{-1} when it reaches the peg.
Find the distance that the block has travelled when it reaches the peg. (3)

(Total 11 marks)

Q7.

A particle, of mass 4 kg, is suspended in equilibrium by two light strings, AP and BP . The string AP makes an angle of 30° to the horizontal and the other string, BP , is horizontal, as shown in the diagram.

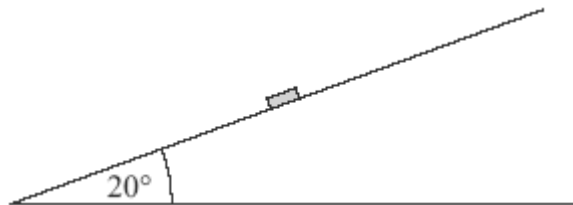


- (a) Draw and label a diagram to show the forces acting on the particle. (1)
- (b) Show that the tension in the string AP is 78.4 N. (3)
- (c) Find the tension in the horizontal string BP . (2)

(Total 6 marks)

Q8.

A puck, of mass 0.2 kg, is placed on a slope inclined at 20° above the horizontal, as shown in the diagram.



The puck is hit so that initially it moves with a velocity of 4 m s^{-1} directly up the slope.

- (a) A simple model assumes that the surface of the slope is smooth.
- (i) Show that the acceleration of the puck up the slope is -3.35 m s^{-2} , correct to three significant figures. (3)
 - (ii) Find the distance that the puck will travel before it comes to rest. (3)
 - (iii) What will happen to the puck after it comes to rest?
Explain why. (2)
- (b) A revised model assumes that the surface is rough and that the coefficient of friction between the puck and the surface is 0.5.
- (i) Show that the magnitude of the friction force acting on the puck during this motion is 0.921 N, correct to three significant figures. (3)
 - (ii) Find the acceleration of the puck up the slope. (3)
 - (iii) What will happen to the puck after it comes to rest in this case?
Explain why. (2)
- (Total 16 marks)

Q9.

A tractor, of mass 4000 kg, is used to pull a skip, of mass 1000 kg, over a rough horizontal surface. The tractor is connected to the skip by a rope, which remains taut and horizontal throughout the motion, as shown in the diagram.



Assume that only **two** horizontal forces act on the tractor. One is a driving force, which has magnitude P newtons and acts in the direction of motion. The other is the tension in

the rope.

The coefficient of friction between the skip and the ground is 0.4.

The tractor and the skip accelerate at 0.8 m s^{-2} .

- (a) Show that the magnitude of the friction force acting on the skip is 3920 N. (2)
- (b) Show that $P = 7920$. (3)
- (c) Find the tension in the rope. (3)
- (d) Suppose that, during the motion, the rope is **not** horizontal, but inclined at a small angle to the horizontal, with the higher end of the rope attached to the tractor, as shown in the diagram.



How would the magnitude of the friction force acting on the skip differ from that found in part (a)?

Explain why.

(2)
(Total 10 marks)

Q10.

Two particles, A and B , are connected by a light inextensible string, which passes over a smooth peg. Particle A is on a rough horizontal surface and has mass 3 kg. Particle B hangs freely, as shown in the diagram, and has mass 2 kg. The coefficient of friction between A and the horizontal surface is μ .



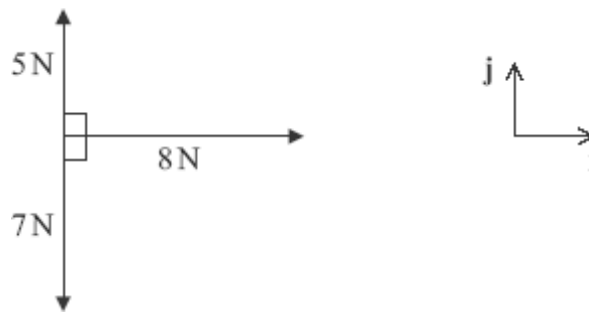
The particles are released from rest and move with a constant acceleration of magnitude 0.9 m s^{-2} .

- (a) Find the tension in the string. (3)
- (b) Draw and label a diagram to show the forces acting on particle A .

- (1)
- (c) Calculate the magnitude of the normal reaction force acting on A . (1)
- (d) Find the magnitude of the friction force that acts on A . (2)
- (e) Find μ . (2)
- (Total 9 marks)

Q11.

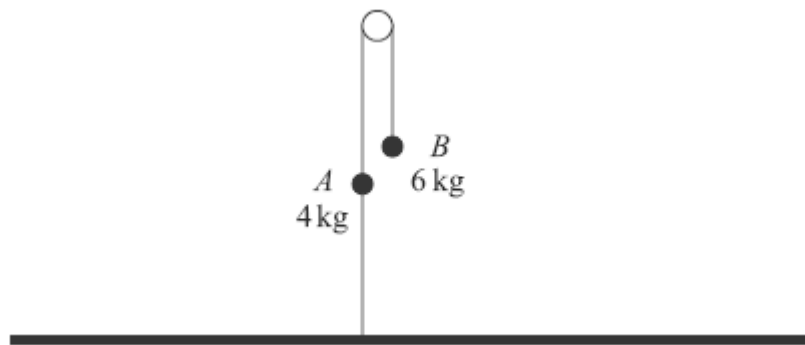
The diagram shows three forces and the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$, which all lie in the same plane.



- (a) Express the resultant of the three forces in terms of $\mathbf{i}$ and $\mathbf{j}$. (2)
- (b) Find the magnitude of the resultant force. (2)
- (c) Draw a diagram to show the direction of the resultant force, and find the angle that it makes with the unit vector $\mathbf{i}$. (3)
- (Total 7 marks)

Q12.

Two particles, A and B , have masses 4 kg and 6 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg. A second light inextensible string is attached to A . The other end of this string is attached to the ground directly below A . The system remains at rest, as shown in the diagram.



- (a) (i) Write down the tension in the string connecting A and B . (1)
- (ii) Find the tension in the string connecting A to the ground. (3)
- (b) The string connecting particle A to the ground is cut. Find the acceleration of A after the string has been cut. (5)
- (Total 9 marks)**

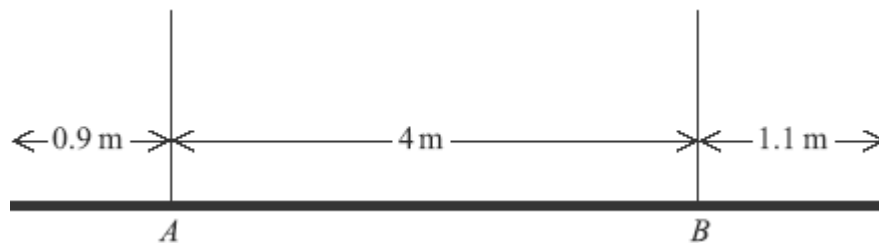
Q13.

A block, of mass 5 kg, slides down a rough plane inclined at 40° to the horizontal. When modelling the motion of the block, assume that there is no air resistance acting on it.

- (a) Draw and label a diagram to show the forces acting on the block. (1)
- (b) Show that the magnitude of the normal reaction force acting on the block is 37.5 N, correct to three significant figures. (2)
- (c) Given that the acceleration of the block is 0.8 m s^{-2} , find the coefficient of friction between the block and the plane. (6)
- (d) In reality, air resistance does act on the block. State how this would change your value for the coefficient of friction and explain why. (2)
- (Total 11 marks)**

Q14.

A uniform plank, of length 6 metres, has mass 40 kg. The plank is held in equilibrium in a horizontal position by two vertical ropes attached to the plank at A and B , as shown in the diagram.



- (a) Draw a diagram to show the forces acting on the plank. (1)
 - (b) Show that the tension in the rope attached to the plank at B is $21g$ N. (3)
 - (c) Find the tension in the rope that is attached to the plank at A. (2)
 - (d) State where in your solution you have used the fact that the plank is uniform. (1)
- (Total 7 marks)**

Q15.

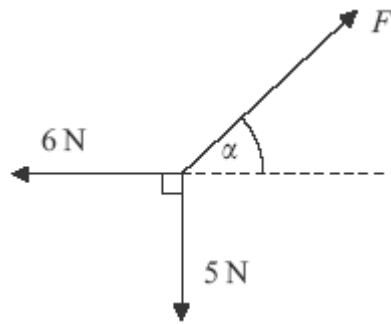
A car, of mass 1500 kg, is towing a caravan, of mass 900 kg, along a straight horizontal road. The caravan is connected to the car by a horizontal tow bar. Resistance forces of magnitudes 400 N and 800 N act on the car and caravan respectively. The acceleration of the car and caravan is 0.8 m s^{-2} .



- (a) Show that the magnitude of the force that the car exerts on the caravan is 1520 N. (3)
 - (b) Find the magnitude of the driving force produced by the car's engine. (3)
- (Total 6 marks)**

Q16.

The diagram shows three forces which act in the same plane and are in equilibrium.



(a) Find F .

(3)

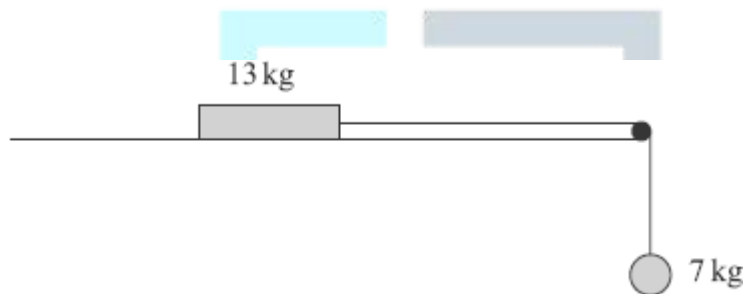
(b) Find α .

(3)

(Total 6 marks)

Q17.

The diagram shows a block, of mass 13 kg, on a rough horizontal surface. It is attached by a string that passes over a smooth peg to a sphere of mass 7 kg, as shown in the diagram.



The system is released from rest, and after 4 seconds the block and the sphere both have speed 6 m s^{-1} , and the block has **not** reached the peg.

(a) State **two** assumptions that you should make about the string in order to model the motion of the sphere and the block.

(2)

(b) Show that the acceleration of the sphere is 1.5 m s^{-2} .

(2)

(c) Find the tension in the string.

(3)

(d) Find the coefficient of friction between the block and the surface.

(6)

(Total 13 marks)

Q18.

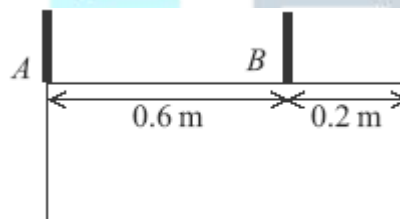
A trolley, of mass 100 kg, rolls at a constant speed along a straight line down a slope inclined at an angle of 4° to the horizontal.

Assume that a constant resistance force, of magnitude P newtons, acts on the trolley as it moves. Model the trolley as a particle.

- (a) Draw a diagram to show the forces acting on the trolley. (1)
- (b) Show that $P = 68.4$ N, correct to three significant figures. (3)
- (c) (i) Find the acceleration of the trolley if it rolls down a slope inclined at 5° to the horizontal and experiences the same constant force of magnitude P that you found in part (b). (4)
- (ii) Make one criticism of the assumption that the resistance force on the trolley is constant. (1)
- (Total 9 marks)**

Q19.

A hotel sign consists of a uniform rectangular lamina of weight W . The sign is suspended in equilibrium in a vertical plane by two vertical light chains attached to the sign at the points A and B , as shown in the diagram. The edge containing A and B is horizontal.



The tensions in the chains attached at A and B are T_A and T_B respectively.

- (a) Draw a diagram to show the forces acting on the sign. (1)
- (b) Find T_A and T_B in terms of W . (4)
- (c) Explain how you have used the fact that the lamina is uniform in answering part (b). (1)
- (Total 6 marks)**

Q20.

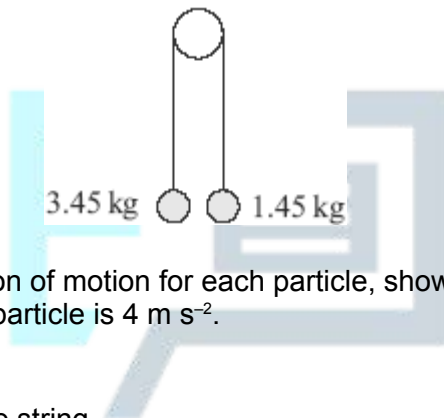
A hot air balloon is at rest on the ground. When the balloon is released, it rises to a height of 320 metres in 80 seconds. The balloon moves under the action of its weight and a vertical lift force. Assume that the balloon has a constant acceleration during this motion.

- (a) Show that the acceleration of the balloon is 0.1 m s^{-2} .

- (3)
- (b) Find the speed of the balloon when it reaches a height of 320 metres. (2)
- (c) The mass of the balloon is 450 kg. Show that the magnitude of the vertical lift force is 4500 N, correct to two significant figures. (3)
- (d) After a while, the vertical lift force is reduced so that the balloon rises at a constant speed. State the magnitude of the vertical lift force when this is the case. (1)
- (Total 9 marks)

Q21.

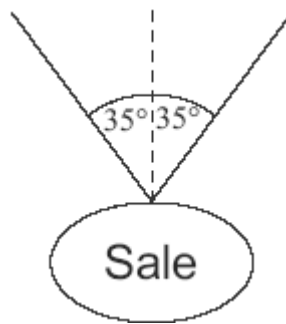
Two particles, of masses 3.45 kg and 1.45 kg, are connected by a light string that passes over a smooth peg. The particles are released from rest with the strings vertical, as shown in the diagram.



- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 4 m s^{-2} . (5)
- (b) Find the tension in the string. (2)
- (c) Initially the particles are at the same level. (3)
- Find the speed of the heavier particle when it is 1 metre lower than the lighter particle. Assume that neither particle hits the floor or the peg.
- (Total 10 marks)

Q22.

A sign, of mass 2 kg, is suspended from the ceiling of a supermarket by two light strings. It hangs in equilibrium with each string making an angle of 35° to the vertical, as shown in the diagram. Model the sign as a particle.

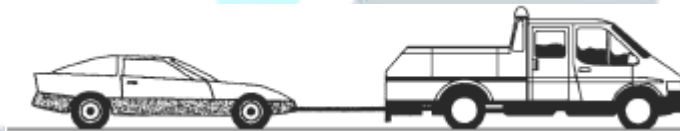


- (a) By resolving forces horizontally, show that the tension is the same in each string. (2)
- (b) Find the tension in each string. (5)
- (c) If the tension in a string exceeds 40 N, the string will break. Find the mass of the heaviest sign that could be suspended as shown in the diagram. (3)

(Total 10 marks)

Q23.

A car, of mass 1200 kg, is connected by a tow rope to a truck, of mass 2800 kg. The truck tows the car in a straight line along a horizontal road. Assume that the tow rope is horizontal. A horizontal driving force of magnitude 3000 N acts on the truck. A horizontal resistance force of magnitude 800 N acts on the car. The car and truck accelerate at 0.4 m s^{-2} .



- (a) Find the tension in the tow rope. (3)
- (b) Show that the magnitude of the horizontal resistance force acting on the truck is 600 N. (4)
- (c) In fact, the tow rope is **not** horizontal. Assume that the resistance forces and the driving force are unchanged.

Is the tension in the tow rope greater or less than in part (a)?

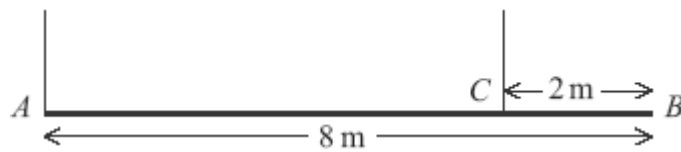
Explain why.

(2)
(Total 9 marks)

Q24.

A uniform plank, AB , is 8 m long and has mass 30 kg. It is supported in equilibrium in a

horizontal position by two vertical inextensible ropes. One of the ropes is attached to the plank at A and the other rope to the point C , where $BC = 2$ m, as shown in the diagram.



Find the tension in each rope.

(Total 5 marks)

Q25.

A uniform plank is 10 m long and has mass 15 kg. It is placed on horizontal ground at the edge of a vertical river bank, so that 2 m of the plank is projecting over the edge, as shown in the diagram below.



- (a) A woman of mass 50 kg stands on the part of the plank which projects over the river.

Find the greatest distance from the river bank at which she can safely stand.

(3)

- (b) The woman wishes to stand safely at the end of the plank which projects over the river.

Find the minimum mass which she should place on the other end of the plank so that she can do this.

(4)

- (c) State how you have used the fact that the plank is uniform in your solution.

(1)

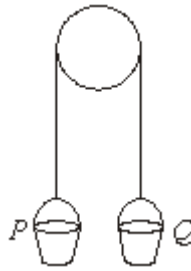
- (d) State one other modelling assumption which you have made.

(1)

(Total 9 marks)

Q26.

A builder ties two identical buckets, P and Q , to the ends of a light inextensible rope. He hangs the rope over a smooth beam so that the buckets hang in equilibrium, as shown in the diagram.



The buckets are each of mass 0.6 kg.

- (a) (i) State the magnitude of the tension in the rope. (1)
- (ii) State the magnitude and direction of the force exerted on the beam by the rope. (2)
- (b) The bucket Q is held at rest while a stone, of mass 0.2 kg, is placed inside it. The system is then released from rest and, in the subsequent motion, bucket Q moves vertically downwards with the stone inside.
- (i) By forming an equation of motion for each bucket, show that the magnitude of the tension in the rope during the motion is 6.72 newtons, correct to three significant figures. (6)
- (ii) State the magnitude of the force exerted on the beam by the rope while the motion takes place. (1)
- (Total 10 marks)

Q27.

A crate is being pulled at constant speed across rough horizontal ground by a rope.

The crate is of weight 100 newtons and the frictional force between the crate and the ground is of magnitude 30 newtons.

The tension in the rope is of magnitude T newtons.

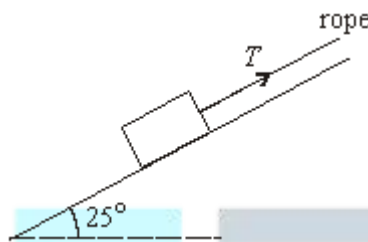


- (a) Draw and label a diagram to show all the forces acting on the crate. (1)
- (b) The coefficient of friction between the crate and the ground is 0.5. Show that the normal reaction force between the crate and the ground is 60 newtons. (2)

- (c) Explain why the horizontal component of the tension in the rope is 30 newtons. (2)
- (d) Find the value of T . (4)
- (e) Find the angle that the rope makes with the horizontal. (3)
- (Total 12 marks)**

Q28.

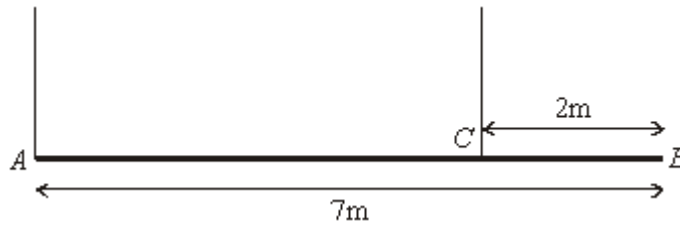
A rough slope is inclined at an angle of 25° to the horizontal. A box of weight 80 newtons is on the slope. A rope is attached to the box and is parallel to the slope. The tension in the rope is of magnitude T newtons. The diagram shows the slope, the box and the rope.



- (a) The box is held in equilibrium by the rope.
- (i) Show that the normal reaction force between the box and the slope is 72.5 newtons, correct to three significant figures. (3)
- (ii) The coefficient of friction between the box and the slope is 0.32. Find the magnitude of the maximum value of the frictional force which can act on the box. (2)
- (iii) Find the least possible tension in the rope to prevent the box from moving down the slope. (4)
- (iv) Find the greatest possible tension in the rope. (3)
- (v) Show that the mass of the box is approximately 8.16 kg. (1)
- (b) The rope is now released and the box slides down the slope. Find the acceleration of the box. (3)
- (Total 16 marks)**

Q29.

A uniform beam, AB , has mass 20 kg and length 7 metres. A rope is attached to the beam at A . A second rope is attached to the beam at the point C , which is 2 metres from B . Both of the ropes are vertical. The beam is in equilibrium in a horizontal position, as shown in the diagram.

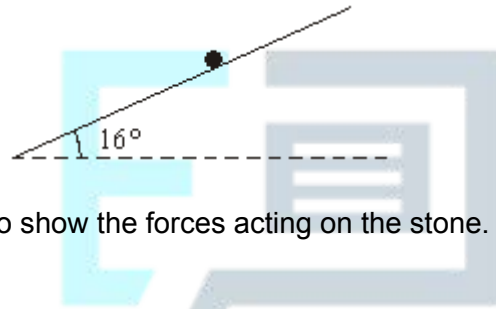


Find the tensions in the two ropes.

(Total 6 marks)

Q30.

A stone rests in equilibrium on a rough plane inclined at an angle of 16° to the horizontal, as shown in the diagram. The mass of the stone is 0.5 kg.

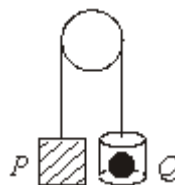


- (a) Draw a diagram to show the forces acting on the stone. (1)
- (b) Show that the magnitude of the frictional force acting on the stone is 1.35 newtons, correct to three significant figures. (3)
- (c) Find the magnitude of the normal reaction force between the stone and the plane. (2)
- (d) Hence find an inequality for the value of μ , the coefficient of friction between the stone and the plane. (2)

(Total 8 marks)

Q31.

A block P is attached to a can Q by a light inextensible string. The string hangs over a smooth peg so that P and Q hang freely, as shown in the diagram.



The block P and the can Q each has mass 0.2 kg . The can Q contains a small stone of mass 0.1 kg . The system is released from rest and the can Q and the stone move vertically downwards.

- (a) By forming two equations of motion, show that the magnitude of the acceleration of P and Q is 1.96 m s^{-2} .

(5)

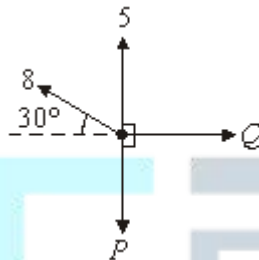
- (b) Find the magnitude of the reaction force between the can and the stone.

(3)

(Total 8 marks)

Q32.

A particle is in equilibrium under the action of four horizontal forces of magnitudes 5 newtons, 8 newtons, P newtons and Q newtons, as shown in the diagram.



- (a) Show that $P = 9$.

(3)

- (b) Find the value of Q .

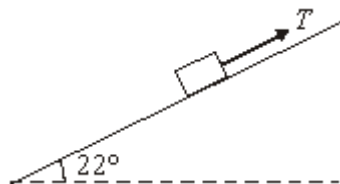
(2)

(Total 5 marks)

Q33.

A block is being pulled up a rough plane inclined at an angle of 22° to the horizontal by a rope parallel to the plane, as shown in the diagram.

The mass of the block is 0.7 kg , and the tension in the rope is T newtons.



- (a) Draw a diagram to show the forces acting on the block.

(1)

- (b) Show that the normal reaction force between the block and the plane has magnitude 6.36 newtons, correct to three significant figures.

(3)

- (c) The coefficient of friction between the block and the plane is 0.25. Find the magnitude of the frictional force acting on the block during its motion.

(2)

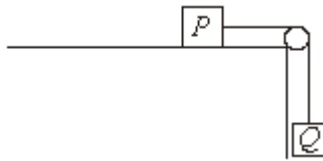
- (d) The tension in the rope is 5.6 newtons. Find the acceleration of the block.

(4)

(Total 10 marks)

Q34.

A small block P is attached to another small block Q by a light inextensible string. The block P rests on a rough horizontal surface and the string hangs over a smooth peg so that Q hangs freely, as shown in the diagram.



The block P has mass 0.4 kg and the coefficient of friction between P and the surface is 0.5.

The block Q has mass 0.3 kg.

The system is released from rest and Q moves vertically downwards.

- (a) (i) Draw a diagram to show the forces acting on P .

(1)

- (ii) Show that the frictional force between P and the surface has magnitude 1.96 newtons.

(2)

- (b) By forming an equation of motion for each block, show that the magnitude of the acceleration of each block is 1.4 m s^{-2} .

(5)

- (c) Find the speed of the blocks after 3 seconds of motion.

(2)

- (d) After 3 seconds of motion, the string breaks. The blocks continue to move. Comment on how the speed of each block will change in the subsequent motion. For each block, give a reason for your answer.

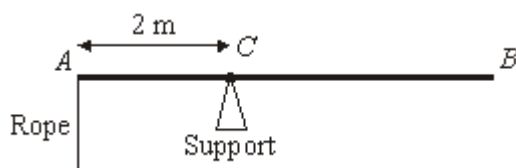
(4)

(Total 14 marks)

Q35.

The diagram shows a uniform rod, AB , of mass 10 kg and length 5 metres. The rod is

held in equilibrium in a horizontal position, by a support at C and a light vertical rope attached to A , where AC is 2 metres.



- (a) Draw and label a diagram to show the forces acting on the rod. (1)
- (b) Show that the tension in the rope is 24.5 N. (3)
- (c) A package of mass m kg is suspended from B . The tension in the rope has to be doubled to maintain equilibrium.
- (i) Find m . (4)
- (ii) Find the magnitude of the force exerted on the rod by the support. (3)
- (d) Explain how you have used the fact that the rod is uniform in your solution. (1)

(Total 12 marks)